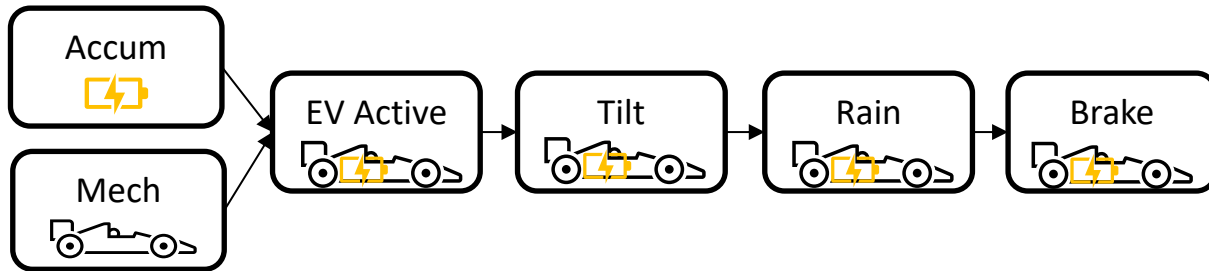


**IMPORTANT**

PRESENT THE VEHICLE FOR INSPECTION IN THE FOLLOWING ORDER:



THIS FORM MUST STAY WITH THE CAR UNTIL THAT SPECIFIC PART OF INSPECTION HAS BEEN COMPLETED

NOTE - IF THERE IS A CONFLICT BETWEEN THIS FORM AND THE RULES, THE RULES PREVAIL

**ACCUMULATOR INSPECTION**

|                                   |                                                                                                                                                                                                                                    |               |  |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--|
| INSPECTOR(s):                     |                                                                                                                                                                                                                                    | UNIVERSITY:   |  |
|                                   | ESF Accepted                                                                                                                                                                                                                       |               |  |
| <b>ACCUMULATOR</b>                |                                                                                                                                                                                                                                    |               |  |
| Accumulator Labels                | Accumulator housing must be labeled with "High Voltage Always Energized" and  . Visible from all angles.                                          | Visible check |  |
|                                   | Accumulator labeled with team number visible from top of accumulator.                                                                                                                                                              | Visible check |  |
| Separation on self-developed PCBs | GLV and TS circuits have at least the spacing specified in EV.6.5.7.                                                                                                                                                               | Visible check |  |
| HV Path                           | Bolted connections in the high current path must have a positive locking mechanism. <b>Lock washers and thread locking compound are NOT allowed.</b> Patch bolts allowed into blind OEM components. (EV.6.4.3)                     | Visible check |  |
|                                   | TS and GLV wiring must be separated by a insulating barrier or distance specified in EV.6.5.5                                                                                                                                      | Visible check |  |
|                                   | Soldering is not allowed in the high current path.                                                                                                                                                                                 | Visible check |  |
| Internals                         | The poles of the accumulator stack(s) and cells must be insulated against the inner wall of the accumulator container if the container is made of electrically conductive material.                                                | Visible check |  |
|                                   | Every accumulator container must contain at least one fuse in the high current path.                                                                                                                                               | Visible check |  |
|                                   | No always energized TS wires leaving accumulator container (All TS wires leaving accumulator disconnected by an IR).                                                                                                               | Visible check |  |
|                                   | Branch circuits are fused within 150mm of source in <b>both</b> positive and negative.                                                                                                                                             | Visible check |  |
|                                   | Fuse and IRs must be separated from the rest of the enclosure by an electrically insulating fireproof material.                                                                                                                    | Visible check |  |
| Maintenance Plugs                 | Maintenance plugs can be removed without tools or further disassembly.                                                                                                                                                             | Visible check |  |
|                                   | Surfaces of the maintenance plugs must be non-conductive except as required to make the electrical connection.                                                                                                                     | Visible check |  |
|                                   | Maintenance plugs must be positively locked when installed.                                                                                                                                                                        | Visible check |  |
|                                   | Maintenance plugs cannot be incorrectly installed.                                                                                                                                                                                 | Visible check |  |
| Internals - Cell stacks           | Each stack has to be separated by the use of an electrically insulating and fire resistant materials towards other stacks in the container and on top of the stack. <b>Air is not a suitable insulation material in this case.</b> | Visible check |  |
| Indicator Light                   | Each container must have an indicator showing that voltages greater than 60V DC are present outside of the container.                                                                                                              | Visible check |  |
|                                   | Indicator must function with accumulator removed from vehicle.                                                                                                                                                                     | Visible check |  |
|                                   | Indicator labeled <b>High Voltage Present</b> .                                                                                                                                                                                    | Visible check |  |
| Accumulator Container Connectors  | All Tractive System connectors outside of an enclosure must include an interlock.                                                                                                                                                  | Visible check |  |
| Chargers                          | Charger connector must incorporate an interlock such that the connectors only become live if is correctly connected.                                                                                                               | Visible check |  |
|                                   | HV charging leads must be orange.                                                                                                                                                                                                  | Visible check |  |
|                                   | Charging leads fused if necessary (accumulator end).                                                                                                                                                                               | Visible check |  |
|                                   | HV+ and HV- shrouded red banana jack available when charging                                                                                                                                                                       | Visible check |  |

**!!TEST AT HIGH VOLTAGE!!**  
**Pack removed on charging cart**

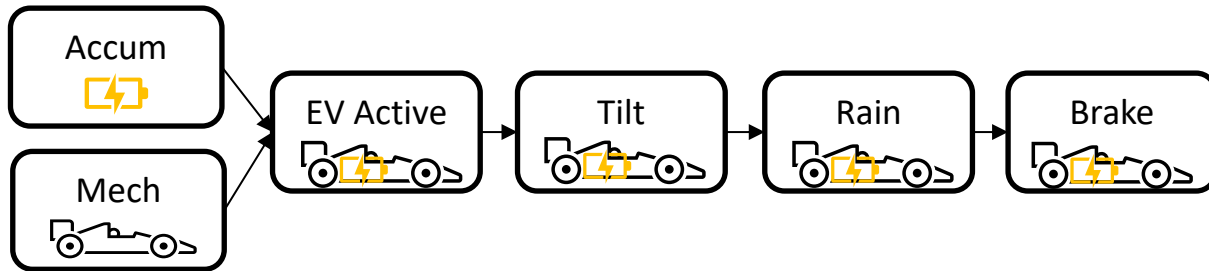
|                       |                                                                                                                                                                                                                                                            |                                                             |  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--|
| Grounding             | Measure resistance between conductive surfaces and ground pin of charger AC connector (mOhm @ 1A), must be < 300mOhm.                                                                                                                                      | _____ mΩ                                                    |  |
| Accumulator Indicator | Accumulator Indicator indicates if voltage above 60VDC is present outside of the container.                                                                                                                                                                | Visible check                                               |  |
| Charging              | Emergency stop button on charger stops charging                                                                                                                                                                                                            | TS voltage must decrease below 60VDC in 5 sec               |  |
|                       | IMD must open shutdown circuit in 30 seconds when isolation fault is present.<br>1. Activate Tractive System<br>2. Connect IMD test box between TSMP and GLVS Ground.<br>3. IMD must trip in 30 seconds and TS voltage must decrease below 60VDC in 5 sec. | Perform Test                                                |  |
|                       | When charging, the BMS must be live and must be able to turn off the charger in the event that a fault is detected.                                                                                                                                        | Set vehicle to charge. Team must demonstrate BMS is active. |  |

APPROVED BY:

DATE/TIME:

**IMPORTANT**

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| DEMONSTRATION                                   |                                                                                                                                                                                                                                        |                   |  |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--|
| UNIVERSITY:                                     |                                                                                                                                                                                                                                        | INSPECTOR(s):     |  |
| Energy Meter                                    | Energy meter installed in vehicle                                                                                                                                                                                                      | Confirm with team |  |
| TS Components                                   | No TS components or wiring below frame.                                                                                                                                                                                                | Visible check     |  |
| Measurements                                    |                                                                                                                                                                                                                                        |                   |  |
| Energy Meter                                    | Energy meter installed in vehicle                                                                                                                                                                                                      | Visible check     |  |
| GLVS ground                                     | Measure GLVS gnd to conductive surface <100mm from a TS component, must be < 300 mOhm @ 1A or 5 Ohm for carbon fiber.                                                                                                                  | _____ mΩ          |  |
| Discharge Circuit and Body Protection Resistors | 1. Measure resistance between TSMP HV- and TSMP HV+<br>$V_{max} < 200V \Rightarrow 10k + \text{discharge}$<br>$200V < V_{max} < 400V \Rightarrow 20k + \text{discharge}$<br>$400V < V_{max} < 600V \Rightarrow 30k + \text{discharge}$ | _____ kΩ          |  |
| Insulation Measurement Test                     | Measure isolation between TSMP and chassis ground. Choose next voltage level above TS voltage (250V or 500V)<br>$R_{iso} \geq 500 * TS \text{ Voltage} + BPR$                                                                          | HV+ _____ MΩ      |  |
|                                                 |                                                                                                                                                                                                                                        | HV- _____ MΩ      |  |

**CAR MUST BE JACKED UP WITH DRIVEN WHEELS REMOVED.**

| !!TEST AT HIGH VOLTAGE!!                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                              |  |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--|
| IMD and Tractive System Status Indicator                                      | IMD indicator light inside the cockpit must be marked with "IMD".                                                                                                                                                                                                                                                                                                                                                                                                                  | Visible check                                                                                                                                |  |
|                                                                               | IMD must open shutdown circuit in 30 seconds when isolation fault is present.<br>1. Activate Tractive System. <b>IMD indicator off, TSSI is green</b><br>2. Connect IMD test box between TSMP and GLVS Ground.<br>3. IMD must trip in 30 seconds and TS voltage must decrease below 60VDC in 5 sec. <b>IMD Indicator is on (red). TSSI is flashing red.</b><br>4. Remove IMD test fault and wait 30 seconds. <b>TS stays off, IMD indicator stays on, TSSI stays flashing red.</b> | Test HV+                                                                                                                                     |  |
|                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Test HV-                                                                                                                                     |  |
| Energy Meter                                                                  | Energy Meter is wired up to GLV and TS connections properly.                                                                                                                                                                                                                                                                                                                                                                                                                       | Visual Check of status LED.                                                                                                                  |  |
| master switches, shutdown buttons and brake-over-travel-switch and interlocks | All switches on --> TS Master switch off                                                                                                                                                                                                                                                                                                                                                                                                                                           | TS voltage must decrease below 60VDC in 5 sec<br><br>Allow team to set pace of tests to prevent overheating precharge / discharge resistors. |  |
|                                                                               | All switches on --> GLV Master switch off                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                              |  |
|                                                                               | All switches on --> left shutdown button off                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                              |  |
|                                                                               | All switches on --> right shutdown button off                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                              |  |
|                                                                               | All switches on --> Cockpit shutdown button off                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                              |  |
|                                                                               | All switches on --> brake-over-travel-switch off                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                              |  |
|                                                                               | All switches on --> Open Interlock(s) of outboard wheel motor (if applicable)                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                              |  |
| OK To Energize Sticker                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                              |  |

|                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                           |  |
|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--|
| Inertia switch                                                                                                                      | Unmount inertia switch. Activate TS and measure HV voltage. Shake the switch and check if TS is shutdown.                                                                                                                                                                                                                                                                                                                                               | TS voltage must decrease below 60VDC in 5 sec                                                                             |  |
| Ready To Move Light                                                                                                                 | The RTML must be flashing amber when GLV turned on and the voltage outside of accumulator container exceeds 60V DC.                                                                                                                                                                                                                                                                                                                                     | Visible check                                                                                                             |  |
|                                                                                                                                     | The RTML must comply with DOT FMVSS 108.                                                                                                                                                                                                                                                                                                                                                                                                                | Visible check                                                                                                             |  |
| Ready-To-Drive Mode                                                                                                                 | Additional actions are required to set the car to ready-to-drive mode<br>1. Enable TS<br>2. Press Accelerator, <b>motors SHOULD NOT spin</b><br>3. Press brake and start button<br>4. Press Accelerator, <b>motors SHOULD spin</b><br>5. Press shutdown button<br>6. Release shutdown button and enable TS<br>7. Press Accelerator, <b>motors SHOULD NOT spin</b><br>8. Press brake and start button<br>9. Press Accelerator, <b>motors SHOULD spin</b> | The team must demonstrate how the car is set to Ready-To-Drive mode by the driver (pressing the brake pedal is mandatory) |  |
| Ready-To-Drive-Sound Test                                                                                                           | The car must make a characteristic sound, once but not continuous, for at least 1 second and a maximum of 3 seconds when it is ready to drive. The sound level must be a minimum of 80dBA, fast weighting, in a radius of 2m around the car. The used sound must be easily recognizable.                                                                                                                                                                | Check for Ready-To-Drive sound when team starts vehicle.                                                                  |  |
| Brake Light                                                                                                                         | Press brake pedal.                                                                                                                                                                                                                                                                                                                                                                                                                                      | Check for sufficient brightness for visible activation in bright sunlight                                                 |  |
| APPS / Brake Pedal Plausibility Check                                                                                               | Torque production must stop when accelerator and brake pedal are pressed simultaneously<br>1. Press accelerator to show that axle turns.<br>2. With accelerator > 25%, press brake pedal. <b>Axle must stop</b><br>3. Keeping accelerator >25%, release brake pedal. <b>Axle must remain stopped.</b><br>4. Slowly release accelerator, <b>axle may turn again once &lt; 5% pedal position.</b>                                                         | Team demonstration per procedure                                                                                          |  |
| APPS Implausibility Check                                                                                                           | Torque production must stop if APPS are implausible.<br>1. Press accelerator to show that axle turns.<br>2. Unplug all but 1 APPS.<br>3. Press accelerator, <b>axle must not turn.</b><br><i>Repeat for each APPS sensor</i>                                                                                                                                                                                                                            | Team demonstration per procedure                                                                                          |  |
| Brake System Plausibility Device                                                                                                    | 1. Enable TS<br>2. Press and hold Brake (may require driver in car)<br>3. Turn up simulated current (5kW at nominal voltage)<br>4. <b>TS must shut off</b> in 0.5 sec<br>5. Release brake and turn off simulated current. <b>The TS may not automatically return to active state.</b> The driver must not be able to reactive the tractive-system.                                                                                                      | Team demonstration per procedure                                                                                          |  |
| Energy Meter                                                                                                                        | Energy Meter functional. Download data from active tests.<br>Verify Voltage, Current and Temperature are reasonable and functional.                                                                                                                                                                                                                                                                                                                     | Data verified                                                                                                             |  |
| <p align="center"><b>!! Test at High Voltages Completed !!</b></p> <p align="center"><b>TRACTIVE SYSTEM HAS TO BE SHUT-OFF!</b></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                           |  |
| APPROVED BY:                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DATE/TIME:                                                                                                                |  |